

Name: _____



Statistical problems - data collection

Task A: Choosing what data to collect

- 1) A supermarket manager collects data to investigate ways to save money, increase sales and improve the customers' experience.

Match the data with the most appropriate tools for collecting it.

customers' opinions about a product that is for sale

questionnaire

number of customers visiting at each time of the day

stopwatch

weight of the vegetables which are left unpurchased

tally chart

time that customers spend queuing at the checkout

weighing scales

Task B: Choosing where to collect the data from

1) Jun wants to find out the average time people take to run a 100 metre race in his school. He assists in a PE lesson for a class in the youngest year group by timing how long each pupil takes to complete the race. He then calculates the mean time.

a) What is the **population** in Jun's investigation?

b) Who does Jun use in his **sample**?

c) Why could Jun's data be **biased**?

d) Propose an alternative way for Jun to collect his data.

2) Laura wants to investigate how much time adults in the UK spend exercising, per week. She designs a questionnaire about people's exercise habits. She goes to her local gym and asks people to complete her questionnaire.

a) What is the **population** in Laura's investigation?

b) Who does Laura use in her **sample**?

c) Why could Laura's data be **biased**?

d) Propose an alternative way for Laura to collect data using her questionnaire.

